**UNITED STATES OF AMERICA**
**BEFORE THE**
**FEDERAL ENERGY REGULATORY COMMISSION**

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| **Ensuring the Timely and Orderly Interconnection of Large Loads** | **Docket No. RM26-4-000** |

**COMMENTS OF DAVID GARDINER AND ASSOCIATES ON MITIGATING LARGE LOAD DEMAND IMPACTS**

Pursuant to Federal Energy Regulatory Commission's ("FERC") Notice Inviting Comments[1] on the Secretary of Energy's advance notice of proposed rulemaking ("Proposed ANOPR"),[2] David Gardiner and Associates ("DGA") is pleased to provide the following initial comments.

FERC's jurisdiction over wholesale rates includes the responsibility to ensure customers are "protect[ed] 'against excessive prices' and [to] ensure effective transmission of electric power."[3] The new large loads present important economic and technological opportunities for our nation; however, due to their unprecedented size, these loads are having, and will continue to have, a direct impact on resource and infrastructure adequacy across the nation which have corresponding impacts on wholesale rates. In exploring solutions that will help facilitate timely

[1] *Interconnection of Large Loads to the Interstate Transmission System*, Notice Inviting Comments, Docket No. RM26-4-000, eLibrary No. 20251027-3056 (2025).

[2] Secretary of Energy, Letter to FERC Commissioners re Initiating Rulemaking Procedures and Proposal Regarding the Interconnection of Large Loads Pursuant to the Secretary's Authority Under Section 403 of the Department of Energy Organization Act (Oct. 23, 2025).

[3] *FERC v. Electric Power Supply Ass'n*, 577 US 260, 136 S.Ct. 760, 781 (2016)(citing *Pennsylvania Water & Power Co. v. FPC,* 343 U.S. 414, 418 (1952); *Gulf States Util. Co. v. FPC,* 411 U.S. 747, 758 (1973)).

and cost-effective interconnection of large loads, DGA encourages FERC to consider a broad range of options that are aimed at both: (1) adding needed infrastructure to the grid—including by facilitating cost-effective transmission development to interconnect additional generation capacity in a timely manner, and (2) mitigating the total load requirements to temper additional infrastructure needs—including by improving the accuracy of load forecasting and by reducing overall demand of individual facilities.

Many commenters in this docket are addressing infrastructure development and load forecasting. As such, DGA's comments are limited to the last issue—ways that FERC can support proactive demand reductions—as these actions can have significant impacts on large load energy needs. For example, data centers—the largest individual sources of projected load growth—can reduce their facilities electricity demand by 10 to 30% by reusing their waste heat.[4] Such energy efficiency tools provide several benefits that are directly related to the issues raised in this docket, including tempering wholesale rates and facilitating speed-to-power by making efficient use of headroom on the system. FERC should include these tools in its menu of options as it moves forward in addressing large load energy needs.

## I. ABOUT DGA

David Gardiner and Associates is a strategic advisory firm that works with a diverse array of clients—including foundations, companies, trade associations, multi-stakeholder organizations, and non-profit organizations—to advance policy and technical solutions that are designed to achieve transformational decarbonization of our nation's energy systems while

[4] Gardiner, David et al., *Data Center Heat Reuse: The Opportunity for States*, at 1-2 (Sept. 2025)("DGA Report").

supporting strong economic development and affordable energy rates. These comments are submitted by DGA on its own behalf, and do not necessarily reflect the views of any of our clients.

## II. COMMENTS

### *A. Large loads are requiring unprecedented levels of energy that have an impact on wholesale rates.*

As Secretary Wright correctly notes in the Proposed ANOPR, the "United States electricity demand is expected to grow at an extraordinary pace, due, in large part, to the rapid growth of large loads. . . most notably data centers."[5] Indeed, the North American Electric Reliability Corporation ("NERC"), found that summer peak demand is expected to rise by 15% over the next 10 years, and winter peak demand is expected to rise by 18%.[6] In so doing it noted that "[t]he size and speed with which data centers (including crypto and AI) can be constructed and connect to the grid presents unique challenges for demand forecasting and planning for system behavior."[7]

Other experts' projections go even further, finding that "the 5-year forecast of utility peak load growth has increased by more than a factor of six, from 24 GW to 166 GW" and that "[e]lectricity use is forecast to increase even more quickly than peak power demand. By 2030, forecasts indicate that total electricity use will increase by 32%."[8] These experts similarly found

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[5] Proposed ANOPR at Background.

[6] North American Electric Reliability Corporation, *2024 Long-Term Reliability Assessment*, at 9 ( July 11, 2025.

[7] *Id.* at 8 (referenced by ANOPR at Background).

[8] Wilson, John et. al., *Power Demand Forecasts Revised Up for Third Year Running, Led by Data Centers*, Grid Strategies (Nov. 2025).

that "data centers are the largest driver of demand and energy growth, accounting for about 55% of demand growth in utility load forecasts over the next five years."[9]

In contrast to traditional load growth, the current increases are concentrated in just a few entities. For example, the Grid Strategies paper states that "sixteen GW-scale data centers with an aggregate demand of nearly 30 GW are scheduled to come online in 2026 and 2027."[10] These sixteen data centers have an aggregate demand that is comparable to the summer peak for all New York State (28.99 GW).[11]

The magnitude of power required by these new load sources are stressing existing resources and impacting wholesale electricity prices as regions need to add additional generation onto already constrained systems. These impacts are not just theoretical. As Virginia Delegate Richard (Rip) Sullivan noted in a recent letter to the PJM Board on the importance of energy efficiency, data shows that "63% of the increase in capacity prices in the 2025/26 auction can be attributed to data centers, translating to $9.3 billion in infrastructure costs that will be passed on to ratepayers."[12]

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[9] *Id.*

[10] *Id.*

[11] New York ISO, *Electric Grid Prepared to Meet 2025 Summer Demand: Reliability Concerns Persist Under Extreme Weather Scenarios* (May 13, 2025).

[12] Delegate Rip Sullivan, *Letter to the PJM Board re "Reduc[ing] Load from Data Centers by Reusing their Heat*, (filed Oct. 31, 2025), attached as Attachment A.

### *B. Data centers can achieve significant demand reductions by implementing energy efficiency measures.*

Status quo energy policies are designed to take load "as it comes" and to plan infrastructure and resource adequacy requirements accordingly. As FERC considers how to best balance its obligations to protect against excessive rates and to ensure effective transmission of power needed to promote economic development, FERC should explore the full spectrum of tools, including tools that mitigate the magnitude of the expected load increases.

Specifically, energy efficiency solutions can have a direct and measurable impact on data center demand and should be considered as part of FERC's solution set. For example, data center electricity demand is concentrated, in large part, towards cooling operations to prevent servers from overheating.[13] Because cooling operations can account for 40% of data center electricity consumption, these facilities can mitigate their overall electricity needs by reusing the data center's heat in other applications. Data center heat reuse can reduce data center electricity demand by 10 to 30%,[14] which translates to "reduc[ing] new power demand by as much as 27 GW, [and] avoiding the construction of as many as 54 new conventional 500 megawatt (MW) power plants."[15]

The low-temperature heat produced by data centers can be effectively recovered and used in several industries, including pharmaceuticals, food and beverage processing, and commercial laundries that depend on hot water preheating. Though data center heat reuse is a proven

[13] U.S. DOE, "DOE Announces $40 Million for More Efficient Cooling for Data Centers," May 9, 2023.

[14] DGA Report at 1-2.

[15] *Id*. Savings are calculated based on data center demand growth projections by Grid Strategies of up to 90 GW. Grid Strategies, *Strategic Industries Surging: Driving US Power Demand*, at 2 (Dec. 2024).

technology that has been widely adopted across Europe, it remains an underutilized tool in the United States.[16]

### *C. FERC and states should incentivize demand reductions.*

While demand reductions can play a key role in resolving resource constraints there is little incentive for data centers to implement these actions unilaterally. FERC has a unique role and opportunity to incentivize demand reductions, and it should use this power to protect existing customers. DGA recommends that such incentives should focus on speed to power rather than market payments for two main reasons. First, unlike traditional demand response and load flexibility tools, which provide real time flexibility to the energy markets, proactive demand reductions provide permanent reductions. Second, time is money and there is more leverage in providing an incentive to interconnect quicker (whether through a new load interconnection process or via the generation interconnection process) than there is in potential future market payments.

## III. CONCLUSION

Managing the significant new large load that is expected will require a whole-of-electricity-sector approach to maintain resource adequacy and keep consumer costs affordable. Demand reductions—such as through reuse of data center waste heat—are an essential piece of this approach. DGA applauds FERC for taking swift action to address the growing concerns around large load growth and looks forward to continuing to engage on this issue.

[16] DGA Report.

Respectfully submitted,

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David Gardiner and Associates

Nov. 21, 2025

**ATTACHMENT A**

COMMONWEALTH OF VIRGINIA
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COMMITTEE ASSIGNMENTS:
FINANCE (VICE CHAIR)
COURTS OF JUSTICE
LABOR AND COMMERCE
RULES

October 31, 2025

The PJM Board of Managers
c/o Mr. David E. Mills, Chair, PJM Board of Managers and
Mr. Manu Asthana, President and CEO, PJM Interconnection
PJM Interconnection, LLC
2750 Monroe Boulevard
Audubon, PA 19043

Re: Reduce Load from Data Centers by Reusing their Heat

Dear Chairman Mills, Mr. Asthana, and the Board of Managers,

As the Delegate representing Fairfax County—part of Virginia's "data center alley"—I am deeply concerned with the impact that data center load is having and will continue to have on my constituents' electricity prices and service reliability. Data centers are an essential component of the Virginia economy, but PJM must take every effort to ensure Virginia residents are not harmed by the proliferation of this industry. Data center load growth is one of the largest factors impacting electricity demand and costs in the PJM region. 63% of the increase in capacity prices in the 2025/26 auction can be attributed to data centers, translating to $9.3 billion in infrastructure costs that will be passed on to ratepayers.[1] The 2025 forecast for PJM's Dominion Zone estimated more than 20 GW of growth from data centers alone by 2037.[2]

While I applaud PJM for taking proactive steps to address data center load growth, I am concerned that one of the easiest solutions to make data centers more efficient and reduce data center demand — the reuse of heat produced at data center heat — has not been raised in the stakeholder conversations. Reuse of data centers' heat reduces these facilities' electricity demand, protecting ratepayers from surging prices while enabling regional economic development. It is essential that we leverage every possible tool to address skyrocketing load

[1] The Institute for Energy Efficiency and Financial Analysis, "Projected data center growth spurs PJM capacity prices by factor of 10," July 30, 2025.
[2] The Institute for Energy Efficiency and Financial Analysis, "Projected data center growth spurs PJM capacity prices by factor of 10," July 30, 2025.

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growth, and many experts have identified energy efficiency and demand reductions as a key tool. PJM should prioritize reusing heat from data centers.

**Heat Reuse is a Readily Available but Underutilized Technology**

Data centers essentially function as heat pumps, with more than 95% of the inputted electricity converted to heat. To prevent the servers from overheating, data centers consume large amounts of electricity—approximately 40% of their total electricity consumption—for cooling.[3]

Reusing data center heat reuse provides an alternative to this electricity-intensive process by transporting the waste heat to neighboring facilities that utilize it. This means the data center needs less electricity for cooling and can cut its demand by 10 to 30%.[4] This could translate to a 1200 to 3600 MW reduction in peak summer demand in the PJM region, greatly alleviating strain on the grid and protecting ratepayers from excess costs for new generation as more data centers come online.[5]

Data center heat reuse is widely deployed in Europe but remains an underutilized tool in the United States. The heat that data centers produce is low-temperature and is well-suited for a variety of applications, including in the pharmaceutical sector, the food and beverage sector, and industries that require hot water pre-heating like commercial laundries. The prevalence of these projects in Europe is largely due to policies that require, incentivize, or enable them, demonstrating the need for policy support in the United States.

**PJM Should Use its Processes to Incentivize Heat Reuse to Reduce Overall Data Center Demand**

Speed to power is the most critical component of data center development, especially as wait times for grid connection lengthen in response to demand surges. PJM should leverage this desire for speed by turning grid connection into a carrot. PJM can prioritize energy efficient data center development by moving those with plans to reuse their heat up in the demand-side interconnection queue. This policy will also deliver the co-benefit of offering greater assurance of what data center demand will materialize, as it can be assumed that projects that secure an

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[3] U.S. Department of Energy (U.S. DOE), "DOE Announces $40 Million for More Efficient Cooling for Data Centers," May 9, 2023.
[4] U.S. DOE, "DOE Announces $40 Million for More Efficient Cooling for Data Centers," May 9, 2023.
[5] Based on the estimated 12,000 MW increase in summer peak power demand from current and planned peak power. Monitoring Analytics, "Analysis of the 2026/2027 Base Residual Auction Part A," October 1, 2025.

offtaker for their heat are further in the development process. As many experts are concerned that PJM's forecasts are inflated, this additional qualification is greatly beneficial.[6]

The PJM Board should integrate data center heat reuse into its interconnection and planning processes. By prioritizing projects that demonstrate a commitment to energy efficiency and heat utilization, PJM can simultaneously reduce electricity demand, shield ratepayers from escalating costs, and enable economic development in the region. The challenges of rapid load growth require equally innovative solutions, and incentivizing data center heat reuse represents a pragmatic and cost-effective approach to ensuring system reliability and affordability for all PJM stakeholders.

I welcome the opportunity to discuss reuse of data center heat further with the PJM Board of Managers and look forward to working with you to address load growth while balancing economic development and affordability for ratepayers.

Best,

Richard C. (Rip) Sullivan, Jr.

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[6] The Institute for Energy Efficiency and Financial Analysis notes that "there are strong reasons to believe PJM's 20-year forecasts for data center growth are inflated" in "Projected data center growth spurs PJM capacity prices by factor of 10," July 30, 2025.